Basics of FE-Analysis
BK10A6400
Lecture 8: Linear 2D plane element, numerical integration

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General

- The code for a cantilever beam solved with Q4 elements

Isoparametric Q4

Basis in local normalized coordinates (often called natural coordinates in FE-literature) for the four-node bilinear Q4 element (8 DOFs):

$$\{1, \xi, \eta, \xi\eta\} \tag{1}$$

The displacement field and geometry are interpolated using the same shape functions:

$$\boldsymbol{u}_h(\xi, \eta) = \sum_{I=1}^{4} N_I(\xi, \eta)\, \boldsymbol{u}_I,$$

$$\boldsymbol{X}_h(\xi, \eta) = \sum_{I=1}^{4} N_I(\xi, \eta)\, \boldsymbol{X}_I.$$

Q4 – strains

The previous strain expressions,

$$\varepsilon_{xx} = \frac{\partial u_h}{\partial X_h} = \frac{\partial u_h}{\partial \xi}\left(\frac{\partial X_h}{\partial \xi}\right)^{-1} \quad ; \quad \varepsilon_{yy} = \frac{\partial v_h}{\partial Y_h} = \frac{\partial v_h}{\partial \eta}\left(\frac{\partial Y_h}{\partial \eta}\right)^{-1}$$

$$\gamma_{xy} = \frac{\partial u_h}{\partial Y_h} + \frac{\partial v_h}{\partial X_h} = \frac{\partial u_h}{\partial \eta}\left(\frac{\partial Y_h}{\partial \eta}\right)^{-1} + \frac{\partial v_h}{\partial \xi}\left(\frac{\partial X_h}{\partial \xi}\right)^{-1}$$

are valid only for the special case in which the element is rectangular and its sides are aligned with the X- and Y-axes. For a general Q4 element, the isoparametric transformations need to use because both X_h and Y_h may depend on both local normalized coordinates ξ and η.

Isoparametric tsrains – Q4

Strains using the infinitesimal strain tensor:

$$\boldsymbol{\varepsilon}_m = \frac{1}{2}\left(\nabla \boldsymbol{u}_h + \nabla \boldsymbol{u}_h^T\right) \tag{2}$$

where

$$\begin{aligned} \nabla \boldsymbol{u}_h &= \frac{\partial \boldsymbol{u}_h}{\partial \boldsymbol{X}_h} \\ &= \frac{\partial \boldsymbol{u}_h}{\partial \boldsymbol{\xi}} \left(\frac{\partial \boldsymbol{X}_h}{\partial \boldsymbol{\xi}}\right)^{-1} \\ &= \frac{\partial \boldsymbol{u}_h}{\partial \boldsymbol{\xi}} \boldsymbol{J}_e^{-1} \end{aligned} \tag{3}$$

A linear relationship between stresses and strains:

$$\boldsymbol{\sigma} = \boldsymbol{D}\boldsymbol{\varepsilon} \tag{4}$$

where strain vector can be written as

$$\boldsymbol{\varepsilon} = \begin{pmatrix} \varepsilon_{xx} \\ \varepsilon_{yy} \\ \gamma_{xy} \end{pmatrix} = \begin{pmatrix} \varepsilon_{m11} \\ \varepsilon_{m22} \\ 2\varepsilon_{m12} \end{pmatrix} \tag{5}$$

It is assumed that there is no stress through plate's thickness. In a case of plane stress i.e. $\sigma_{zz} = 0, \sigma_{13} = 0, \sigma_{23} = 0$, elasticity matrix $\boldsymbol{D}$ is written as:

$$\boldsymbol{D} = \frac{E}{1-\nu^2} \begin{pmatrix} 1 & \nu & 0 \\ \nu & 1 & 0 \\ 0 & 0 & \frac{1-\nu}{2} \end{pmatrix} \tag{6}$$

Strain energy:

$$
\begin{aligned}
U &= \frac{1}{2}\int_V \boldsymbol{\sigma}^T \boldsymbol{\varepsilon}\,\mathrm{d}V \\
&= \frac{1}{2} l_z \int_{\Omega_R} \boldsymbol{\sigma}^T \boldsymbol{\varepsilon}\,\det(\boldsymbol{J}_e)\,\mathrm{d}\boldsymbol{\xi} \\
&= \frac{1}{2} l_z \int_{-1}^{1}\int_{-1}^{1} \boldsymbol{\sigma}^T \boldsymbol{\varepsilon}\,\det(\boldsymbol{J}_e)\,\mathrm{d}\xi\mathrm{d}\eta
\end{aligned}
\tag{7}
$$

where the volume (area in 2D) transformation between reference configuration and initial configuration is taken into account via $\det(\boldsymbol{J}_e)$

Numerical integration

- To find an approximate solution to a definite integral $\int_{-1}^{1} f(x)dx \approx \sum_{i=1}^{nip} f(x_i)w_i$.
- Needed to evaluate area or volume integral to get $\overline{\boldsymbol{K}}$
- So far, we have computed it analytically
- Why to make it numerically?
  - Faster (for complicated element)
  - Analytical solution is not always possible
  - Specific numerical integration schemes are needed to get rid of locking phenomena
- Usually Gaussian quadratures are used because a polynomial of degree $p = 2nip - 1$ can be integrated exactly with nip integration points

$$W_{int} = \frac{1}{2}\int_V \boldsymbol{\sigma}^T \boldsymbol{\varepsilon}\,\mathrm{d}V = \frac{1}{2}\boldsymbol{u}^T \underbrace{\int_V \boldsymbol{B}^T \boldsymbol{D}\boldsymbol{B}\,\mathrm{d}V}_{\overline{\boldsymbol{K}}}\,\boldsymbol{u} \tag{8}$$

$$W_{int} = \frac{1}{2} l_z \int_A \boldsymbol{\sigma}^T \boldsymbol{\varepsilon}\,\mathrm{d}A = \frac{1}{2}\boldsymbol{u}^T\, l_z \int_A \boldsymbol{B}^T \boldsymbol{D}\boldsymbol{B}\,\mathrm{d}A\,\boldsymbol{u} \tag{9}$$

Matlab class assigment: Numerical integration

Use Gauss quadrature integration scheme to estimate function $A(x) = \int_a^b f(x)dx$ where $f(x) = x^2$ and $a = 0, b = 3$.

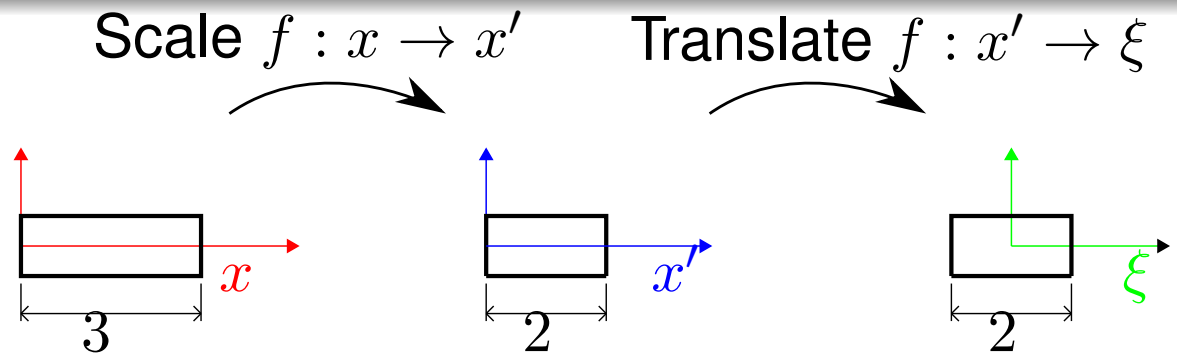
$$A = \int_a^b f(x)dx = \int_0^b x^2 dx \tag{10}$$

Transformation: $x \rightarrow \xi$

$$x = \frac{b}{2}(\xi + 1) \quad ; dx = \frac{b}{2}\, d\xi \tag{11}$$

$$A = \int_{-1}^{1} f(\xi)\frac{b}{2}\, d\xi \approx \sum_{i=1}^{nip} f(\xi_i) w_i \frac{b}{2} \tag{12}$$

Matlab class assigment: transformation $x \rightarrow \xi$ step by step



Scaling $x \rightarrow x'$ (from coordinates $0 \ldots 3 \rightarrow 0 \ldots 2$) :

$$x' = \frac{2}{3}x \quad ; \quad x'(x = 0) = 0\,;\, x'(x = 3) = 2 \tag{13}$$

Translating $x' \rightarrow \xi$ (from coordinates $0 \ldots 2 \rightarrow -1 \ldots 1$) :

$$\xi = x' - 1 \quad ; \quad \xi(x' = 0) = -1\,;\, \xi(x' = 2) = 1 \tag{14}$$

Substituting x', final eq:

$$\xi = \frac{2}{3}x - 1 \quad ; \quad \xi(x = 0) = -1\,;\, \xi(x = 3) = 1 \tag{15}$$

Matlab class assigment: transformation $x \rightarrow \xi$ step by step

Now x can be solved from Eq. (15):

$$x = \frac{3}{2}(\xi + 1) \tag{16}$$

and differential dx:

$$\frac{dx}{d\xi} = \frac{3}{2} \quad \rightarrow \quad dx = \frac{3}{2} d\xi \tag{17}$$

Now the area $A(x)$ can be written

$$A = \int_0^3 x^2 dx = \int_{-1}^{1} \underbrace{\left(\frac{3}{2}(\xi + 1)\right)^2}_{f(\xi)} \underbrace{\frac{3}{2} d\xi}_{dx} \tag{18}$$

Matlab class assigment: Numerical integration

A number of integration points for an accurate solution. The degree of a polynomial of the integrand $p = 2$.

$$nip = \frac{(p+1)}{2} = 1.5 \tag{19}$$

so two integration points are needed. For Gaussian quadrature integration, integration points and weights can be found in mathematics handbooks.

Matlab class assignment: Numerical integration in Matlab

Two point Gauss quadrature formula is exact for polynomials up to degree 2:

```
xiv=[-0.57735026918963,0.57735026918963];
wxiv=[1,1];
```

You need to write function where is your integrand $f(x) = x^2$ and then main code where integration is done:

```
f=0;
for i=1:nip,
   f=f+wxiv(i)*fdxi(xiv(i));
end
```

Numerical integration in 2D and 3D

2D:

$$\int_{-1}^{1}\int_{-1}^{1} f(\xi,\eta)\,\mathrm{d}\xi\mathrm{d}\eta \approx \sum_{j=1}^{nip_\eta}\sum_{i=1}^{nip_\xi} f(\xi_i,\eta_j)w_{\xi_i}w_{\eta_j} \tag{20}$$

3D:

$$\int_{-1}^{1}\int_{-1}^{1}\int_{-1}^{1} f(\xi,\eta,\zeta)\,\mathrm{d}\xi\mathrm{d}\eta \approx \sum_{k=1}^{nip_\zeta}\sum_{j=1}^{nip_\eta}\sum_{i=1}^{nip_\xi} f(\xi_i,\eta_j,\zeta_j)w_{\xi_i}w_{\eta_j}w_{\zeta_k} \tag{21}$$

Shear locking and selective reduced integration

To get rid of shear locking (of bilinear plate element), you should make separated integration for energies related to shear and normal deformations.

$$U = U_n + U_s \tag{22}$$

$$U_n = \frac{1}{2} l_z \int_V \underbrace{\boldsymbol{\sigma}_n^T \boldsymbol{\varepsilon}_n \det(\boldsymbol{J}_e)}_{f_n(\xi,\eta)} \, \mathrm{d}\boldsymbol{\xi} = \frac{1}{2} l_z \int_V \boldsymbol{\varepsilon}_n^T \boldsymbol{D}_n \boldsymbol{\varepsilon}_n \det(\boldsymbol{J}_e) \, \mathrm{d}\boldsymbol{\xi} \tag{23}$$

$$U_s = \frac{1}{2} l_z \int_V \underbrace{\sigma_{xy} \gamma_{xy} \det(\boldsymbol{J}_e)}_{f_s(\xi,\eta)} \, \mathrm{d}\boldsymbol{\xi} = \frac{1}{2} l_z \int_V D_s \gamma_{xy}^2 \det(\boldsymbol{J}_e) \, \mathrm{d}\boldsymbol{\xi} \tag{24}$$

Note that $f_s(\xi, \eta)$ and $f_s(\xi, \eta)$ are integrands and a number of integration points are estimated based on those functions.

Shear locking and selective reduced integration

$$\boldsymbol{\sigma}_n = \begin{pmatrix} \sigma_{11} \\ \sigma_{22} \end{pmatrix} = \begin{pmatrix} \sigma_{xx} \\ \sigma_{yy} \end{pmatrix}$$

$$\boldsymbol{\varepsilon}_n = \begin{pmatrix} \varepsilon_{11} \\ \varepsilon_{22} \end{pmatrix} = \begin{pmatrix} \varepsilon_{xx} \\ \varepsilon_{yy} \end{pmatrix}$$

Lets split material stiffness matrix into shear and normal parts:

$$\boldsymbol{D} = \frac{E}{1-\nu^2} \begin{pmatrix} 1 & \nu & 0 \\ \nu & 1 & 0 \\ 0 & 0 & \frac{1-\nu}{2} \end{pmatrix}$$

$$\boldsymbol{D}_n = \frac{E}{1-\nu^2} \begin{pmatrix} 1 & \nu \\ \nu & 1 \end{pmatrix} \quad ; \quad D_s = \frac{E}{1-\nu^2} \frac{1-\nu}{2}$$

Shear locking and selective reduced integration

One-point quadrature for a shear part and two-points-quadrature for a normal stress part are used to eliminate shear locking phenomena as follows:

$$U = \sum_{i=1}^{2}\sum_{j=1}^{2} f_n(\xi_i, \eta_j) w_i w_j + f_s(\xi_1, \eta_1) w_1$$

where

$$f_n = \frac{1}{2} l_z \boldsymbol{\varepsilon}_n^T \boldsymbol{D}_n \boldsymbol{\varepsilon}_n \det(\boldsymbol{J}_e) \tag{25}$$

$$f_s = \frac{1}{2} l_z D_s \gamma_{xy}^2 \det(\boldsymbol{J}_e) \tag{26}$$

Distributed forces

- Distributed equally on nodes
- Distributed by using shape functions

The variation (i.e virtual work) for the externally applied forces:

$$\delta W_{ext} = \boldsymbol{f}_{ext} \delta \boldsymbol{u}_h = \int_V \boldsymbol{b}^T \boldsymbol{N}_m \, \mathrm{d}V \, \delta \boldsymbol{u} \tag{27}$$

where $\boldsymbol{b}$ is a vector of body forces $[N/m^3]$.

$$\boldsymbol{f}_{ext} = \int_V \boldsymbol{b}^T \boldsymbol{N}_m \, \mathrm{d}V \tag{28}$$

Plane stress and plane strain assumptions

Plane stress

- Suitable for *thin structures* loaded mainly in their plane
- The surfaces normal to the z-direction are free of traction
- Out-of-plane stresses vanish:

$$\sigma_{zz} = \tau_{xz} = \tau_{yz} = 0$$

- The out-of-plane strain ε_{zz} is generally non-zero
- Typical examples: thin plates and sheets loaded in their plane

Plane strain

- Suitable for *long or strongly constrained structures*
- Deformation in the out-of-plane z-direction is prevented
- Out-of-plane strains vanish:

$$\varepsilon_{zz} = \gamma_{xz} = \gamma_{yz} = 0$$

- The out-of-plane stress σ_{zz} is generally non-zero
- Typical examples: dams, tunnels and long structures with nearly constant cross-section

Plane stress vs plane strain

The main difference can be summarized as

Plane stress: $\sigma_{zz} = 0$ | Plane strain: $\varepsilon_{zz} = 0$

Plane stress

- Out-of-plane stress is assumed to vanish
- The structure is free to deform in the z-direction
- Therefore, generally

$$\varepsilon_{zz} \neq 0$$

Plane strain

- Out-of-plane deformation is prevented
- Therefore

$$\varepsilon_{zz} = 0$$

- A non-zero out-of-plane stress σ_{zz} is generally required to enforce this constraint

Constitutive relation

For linear isotropic elasticity,

$$\boldsymbol{\sigma} = \boldsymbol{D}\,\boldsymbol{\varepsilon},$$

where the in-plane stress and strain vectors are

$$\boldsymbol{\sigma} = \begin{bmatrix} \sigma_{xx} \\ \sigma_{yy} \\ \tau_{xy} \end{bmatrix}, \qquad \boldsymbol{\varepsilon} = \begin{bmatrix} \varepsilon_{xx} \\ \varepsilon_{yy} \\ \gamma_{xy} \end{bmatrix}.$$

The constitutive matrix $\boldsymbol{D}$ depends on whether the *plane stress* or *plane strain* assumption is used.

Constitutive matrix: plane stress

For the plane stress assumption,

$$\sigma_{zz} = \tau_{xz} = \tau_{yz} = 0.$$

The constitutive relation is

$$\begin{bmatrix} \sigma_{xx} \\ \sigma_{yy} \\ \tau_{xy} \end{bmatrix} = \boldsymbol{D}_{\text{ps}} \begin{bmatrix} \varepsilon_{xx} \\ \varepsilon_{yy} \\ \gamma_{xy} \end{bmatrix},$$

$$\boldsymbol{D}_{\text{ps}} = \frac{E}{1-\nu^2} \begin{bmatrix} 1 & \nu & 0 \\ \nu & 1 & 0 \\ 0 & 0 & \dfrac{1-\nu}{2} \end{bmatrix}.$$

Although $\sigma_{zz} = 0$, the out-of-plane strain is generally

$$\varepsilon_{zz} \neq 0$$

due to Poisson's effect.

Constitutive matrix: plane strain

For the plane strain assumption,

$$\varepsilon_{zz} = \gamma_{xz} = \gamma_{yz} = 0.$$

The constitutive relation is

$$\begin{bmatrix} \sigma_{xx} \\ \sigma_{yy} \\ \tau_{xy} \end{bmatrix} = \boldsymbol{D}_{\mathrm{pe}} \begin{bmatrix} \varepsilon_{xx} \\ \varepsilon_{yy} \\ \gamma_{xy} \end{bmatrix},$$

where

$$\boldsymbol{D}_{\mathrm{pe}} = \frac{E}{(1+\nu)(1-2\nu)} \begin{bmatrix} 1-\nu & \nu & 0 \\ \nu & 1-\nu & 0 \\ 0 & 0 & \dfrac{1-2\nu}{2} \end{bmatrix}.$$

Although $\varepsilon_{zz} = 0$, the out-of-plane stress is generally

$$\sigma_{zz} \neq 0.$$

Plane stress and plane strain – comparison

	Plane stress	**Plane strain**
Typical structure	Thin	Long / constrained
Assumption	$\sigma_{zz} = 0$	$\varepsilon_{zz} = 0$
Out-of-plane strain	$\varepsilon_{zz} \neq 0$	$\varepsilon_{zz} = 0$
Out-of-plane stress	$\sigma_{zz} = 0$	$\sigma_{zz} \neq 0$
Typical stiffness	More compliant	Stiffer

For the same material properties and in-plane loading, plane strain is generally stiffer than plane stress because deformation in the out-of-plane direction is constrained.